



SAFETY DATA SHEET

CD-234

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : CD-234

Other means of identification : Not applicable

Recommended use : Degreaser

Product dilution information : 0.2 % - 3.1 %

Company :

Chile ECOLAB S.A., Avenida Pedro de Valdivia 3801, Santiago, Chile, (2)-22413300, Telephone: (2)-22381603, Telephone: SAC: 600 241 6600, sac.chile@ecolab.com

Ecolab Argentina S.R.L., Street 11 N° 978, Industrial Park Pilar - Pilar, Buenos Aires, Zip Code: B1629MXA, Telephone: 0230 467 1000

Ecolab Colombia S.A., Avenida Carrera 60 No. 22 – 75 Interior 7., Bogotá, Colômbia, Telephone: 01 8000-120020

ECOLAB Costa Rica SRL, BES Free Zone, El Coyol 7 km west Juan Santamaria Airport, Alajuela, Costa Rica, ZIP Code: Apdo109-4003

Ecolab Mexico, Avenida Industriales 28, Fraccionamiento Industrial - Cuamatla, Cuautitlán Izcalli, Mexico State, Zip Code: 54730, Telephone: (0155) 5864-1800

Quimiproduktos S. de R.L. de C.V., Carretera Miguel Alemán, 990, Colonia La Encarnación - Apodaca, Nuevo León, Mexico, Zip Code: 66633, Telephone: (0181) 47395900

Peru, Calle Santa Isabel # 326 Miraflores. Lima - Perú, Telephone: 511 7176200

Emergency telephone :

Argentina: Ciquime: 0800-222-2933/ 011 4613-1100, National Poison Control Center: 0800 333 0160

Chile: Poison Information Centre CITUC: (+56-2) 2635-3800

Colômbia: CISPROMQUIM® (24 hour service) Phone: 2886012 (Bogotá), Telephone: 018000916012 (Colombia), Telephone: 08001005012 (Venezuela), Telephone: 1800-59-3005 (Ecuador: solo Quito, La sierra, Centro and norte), Telephone: Ecolab USA and Canadá 1-651-222-5352

Costa Rica: For medical emergency information, call 911 and (506) 2436-3900 in Costa Rica. National Poison Control Center in Costa Rica: (506)2223-1028

Mexico: Telephone: 01 800 837 2998 Please request assistance in spanish language

Mexico State: Telephone: 01 800 837 2998

Peru: Telephone: 1-651-222-5352 Please request assistance in spanish language

Issuing date : 30.07.2018

SECTION 2. HAZARDS IDENTIFICATION

Product AS SOLD

Physical and chemical hazards : Not flammable or combustible.

Potential Health Effects

Eyes : Causes serious eye irritation.

Skin : Causes skin irritation. May cause allergic skin reaction.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

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Product AT USE DILUTION

Physical and chemical hazards : Not flammable or combustible.

Potential Health Effects

Eyes : Health injuries are not known or expected under normal use.

Skin : Health injuries are not known or expected under normal use.

Ingestion : Health injuries are not known or expected under normal use.

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Chronic Exposure : Health injuries are not known or expected under normal use.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Product AS SOLD

Pure substance/mixture : Mixture

Chemical name	CAS-No.	Concentration (%)
Limonen	5989-27-5	10 - 30
alcohols, c12-14-secondary, ethoxylated	84133-50-6	5 - 10
Dodecylbenzenesulfonic acid, sodium salt	25155-30-0	5 - 10
Dipropylene glycol methyl ether	34590-94-8	1 - 5
1-methoxy-2-propanol	107-98-2	1 - 5
amines, coco alkyldimethyl, n-oxides	61788-90-7	0.1 - 1

Product AT USE DILUTION

Chemical name	CAS-No.	Concentration (%)
Limonen	5989-27-5	0.1 - 1

SECTION 4. FIRST AID MEASURES

Product AS SOLD

- In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention.
- In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention.
- If swallowed : Rinse mouth. Get medical attention if symptoms occur.
- If inhaled : Get medical attention if symptoms occur.
- Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.
- Notes to physician : Treat symptomatically.
- Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

Product AT USE DILUTION

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In case of eye contact	: Rinse with plenty of water.
In case of skin contact	: Rinse with plenty of water.
If swallowed	: Rinse mouth. Get medical attention if symptoms occur.
If inhaled	: Get medical attention if symptoms occur.

SECTION 5. FIRE-FIGHTING MEASURES

Product AS SOLD

Suitable extinguishing media	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	: None known.
Specific hazards during fire fighting	: Not flammable or combustible.
Hazardous combustion products	: Decomposition products may include the following materials: Carbon oxides Nitrogen oxides (NOx) Sulfur oxides Oxides of phosphorus
Special protective equipment for fire-fighters	: Use personal protective equipment.
Specific extinguishing methods	: Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Product AS SOLD

Personal precautions, protective equipment and emergency procedures	: Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.
Environmental precautions	: Do not allow contact with soil, surface or ground water.
Methods and materials for containment and cleaning up	: Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures	: Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.
Environmental precautions	: Do not allow contact with soil, surface or ground water.
Methods and materials for	: Stop leak if safe to do so. Contain spillage, and then collect with non-

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containment and cleaning up : combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

SECTION 7. HANDLING AND STORAGE

Product AS SOLD

Advice on safe handling : Avoid contact with skin and eyes. Do not get in eyes, on skin, or on clothing. Wash hands thoroughly after handling.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

Storage temperature : 10 °C to 50 °C

Product AT USE DILUTION

Advice on safe handling : Wash hands thoroughly after handling.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Ingredients with workplace control parameters

Ingredients	CAS-No.	Form of exposure	Permissible concentration	Basis
Limonen	5989-27-5	TWA	30 ppm	US WEEL
Dipropylene glycol methyl ether	34590-94-8	CMP	200 ppm	AR OEL
		CMP - CPT	150 ppm	AR OEL
Dipropylene glycol methyl ether	34590-94-8	STEL	150 ppm 909 mg/m3	PE OEL
		TWA	100 ppm 606 mg/m3	PE OEL
Dipropylene glycol methyl ether	34590-94-8	L-8/40	50 ppm 308 mg/m3	VE OEL
Dipropylene glycol methyl ether	34590-94-8	CPT	100 ppm 600 mg/m3	PA OEL
		CCT	150 ppm 900 mg/m3	PA OEL
Dipropylene glycol methyl ether	34590-94-8	VLE-PPT	100 ppm	NOM-010-STPS-2014
		VLE-CT	150 ppm	NOM-010-STPS-2014
Dipropylene glycol methyl ether	34590-94-8	TWA	100 ppm	ACGIH
		STEL	150 ppm	ACGIH
		STEL	150 ppm 900 mg/m3	NIOSH REL
		TWA	100 ppm 600 mg/m3	NIOSH REL
		TWA	100 ppm 600 mg/m3	OSHA Z1
1-methoxy-2-propanol	107-98-2	CMP	100 ppm	AR OEL
		CMP - CPT	150 ppm	AR OEL

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1-methoxy-2-propanol	107-98-2	TWA	100 ppm 369 mg/m3	PE OEL
		STEL	150 ppm 553 mg/m3	PE OEL
1-methoxy-2-propanol	107-98-2	L-8/40	100 ppm	VE OEL
		LB	150 ppm	VE OEL
1-methoxy-2-propanol	107-98-2	CCT	150 ppm 553 mg/m3	PA OEL
		CPT	100 ppm 369 mg/m3	PA OEL
1-methoxy-2-propanol	107-98-2	VLE-PPT	100 ppm	NOM-010-STPS-2014
		VLE-CT	150 ppm	NOM-010-STPS-2014
1-methoxy-2-propanol	107-98-2	TWA	50 ppm	ACGIH
		STEL	100 ppm	ACGIH
		STEL	150 ppm 540 mg/m3	NIOSH REL
		TWA	100 ppm 360 mg/m3	NIOSH REL

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety glasses with side-shields

Hand protection : Wear the following personal protective equipment:
Standard glove type.
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use.
Wash face, hands and any exposed skin thoroughly after handling.

Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Product AS SOLD

Product AT USE DILUTION

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Appearance	: gel	liquid
Color	: amber	amber
Odor	: citrus	slight
pH	: 8.0 - 9.0, (100 %)	7.8 - 8.8
Flash point	: closed cupNot applicable, Sustains combustion	
Odor Threshold	: No data available	
Melting point/freezing point	: No data available	
Initial boiling point and boiling range	: > 100 °C	
Evaporation rate	: No data available	
Flammability (solid, gas)	: No data available	
Upper explosion limit	: No data available	
Lower explosion limit	: No data available	
Vapor pressure	: No data available	
Relative vapor density	: No data available	
Relative density	: 0.963 - 0.993	
Water solubility	: soluble	
Solubility in other solvents	: No data available	
Partition coefficient: n-octanol/water	: No data available	
Autoignition temperature	: No data available	
Thermal decomposition	: No data available	
Viscosity, kinematic	: No data available	
Explosive properties	: No data available	
Oxidizing properties	: No data available	
Molecular weight	: No data available	
VOC	: No data available	

SECTION 10. STABILITY AND REACTIVITY

Product AS SOLD

Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: None known.
Incompatible materials	: None known.
Hazardous decomposition products	: Decomposition products may include the following materials: Carbon oxides Nitrogen oxides (NOx) Sulfur oxides Oxides of phosphorus

SECTION 11. TOXICOLOGICAL INFORMATION

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Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Product AS SOLD

Eyes : Causes serious eye irritation.

Skin : Causes skin irritation. May cause allergic skin reaction.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Eyes : Health injuries are not known or expected under normal use.

Skin : Health injuries are not known or expected under normal use.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact : Redness, Pain, Irritation

Skin contact : Redness, Irritation, Allergic reactions

Ingestion : No symptoms known or expected.

Inhalation : No symptoms known or expected.

Product AT USE DILUTION

Eye contact : No symptoms known or expected.

Skin contact : No symptoms known or expected.

Ingestion : No symptoms known or expected.

Inhalation : No symptoms known or expected.

Toxicity

Product AS SOLD

Product

Acute oral toxicity : No data available

Acute inhalation toxicity : No data available

Acute dermal toxicity : No data available

Skin corrosion/irritation : No data available

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Serious eye damage/eye irritation : No data available

Respiratory or skin sensitization : No data available

Carcinogenicity : No data available

Reproductive effects : No data available

Germ cell mutagenicity : No data available

Teratogenicity : No data available

STOT-single exposure : No data available

STOT-repeated exposure : No data available

Aspiration toxicity : No data available

Ingredients

Acute oral toxicity : Limonen
LD50 Rat: 4,400 mg/kg

alcohols, c12-14-secondary, ethoxylated
LD50 Rat: > 3,000 mg/kg

Dodecylbenzenesulfonic acid, sodium salt
LD50 Rat: 1,086 mg/kg

Dipropylene glycol methyl ether
LD50 Rat: 5,135 mg/kg

1-methoxy-2-propanol
LD50 Rat: > 5,000 mg/kg

Ingredients

Acute inhalation toxicity : 1-methoxy-2-propanol
4 h LC50 Rat: 27.3 mg/l
Test atmosphere: vapor

Ingredients

Acute dermal toxicity : Limonen
LD50 Rabbit: > 5,000 mg/kg

alcohols, c12-14-secondary, ethoxylated
LD50 Rabbit: > 2,000 mg/kg

Dipropylene glycol methyl ether
LD50 Rabbit: 9,510 mg/kg

1-methoxy-2-propanol
LD50 Rabbit: > 13,000 mg/kg

SECTION 12. ECOLOGICAL INFORMATION

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Ecotoxicity

Environmental Effects : Harmful to aquatic life. Toxic to aquatic life with long lasting effects.

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Toxicity to fish : 96 h LC50: 10.5 mg/l

Toxicity to daphnia and other aquatic invertebrates : No data available

Toxicity to algae : No data available

Persistence and degradability

Product AS SOLD

Biodegradable

Bioaccumulative potential

No data available

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. DISPOSAL CONSIDERATIONS

Product AS SOLD

Disposal methods : The product should not be allowed to enter drains, water courses or the soil. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Product AT USE DILUTION

Disposal methods : The product should not be allowed to enter drains, water courses or the soil. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

SECTION 14. TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport

UN number : 3082

Description of the goods : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Limonene)

Class : 9

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Packing group : III
Environmentally hazardous : yes

Air transport (IATA)

Contact Regulatory for air freight eligibility

Sea transport (IMDG/IMO)

UN number : 3082
Description of the goods : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Limonene)
Class : 9
Packing group : III
Marine pollutant : yes

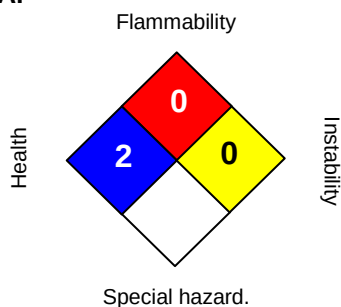
SECTION 15. REGULATORY INFORMATION

For Regulatory information please contact your Ecolab sales representative.

SECTION 16. OTHER INFORMATION

Product AS SOLD

NFPA:



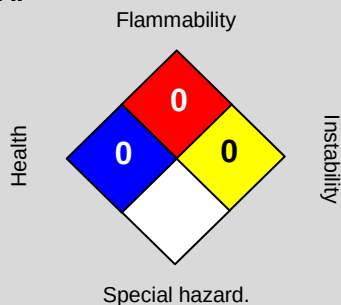
HMIS III:

HEALTH	2*
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Product AT USE DILUTION

NFPA:



HMIS III:

HEALTH	0
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Issuing date : 30.07.2018
Version : 1.0
Prepared by : Regulatory Affairs

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REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the MSDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.